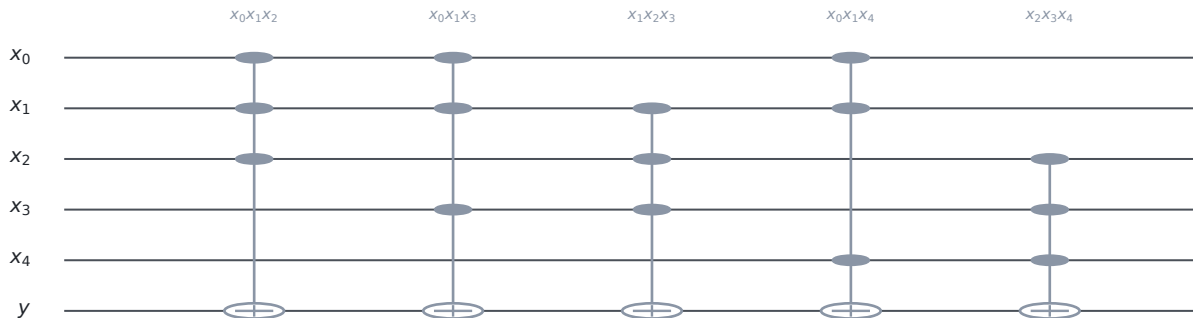


$$f = x_0x_1(x_2 \oplus x_3 \oplus x_4) \oplus x_2x_3(x_1 \oplus x_4) \quad | \quad \text{truth table 0xB008C880}$$

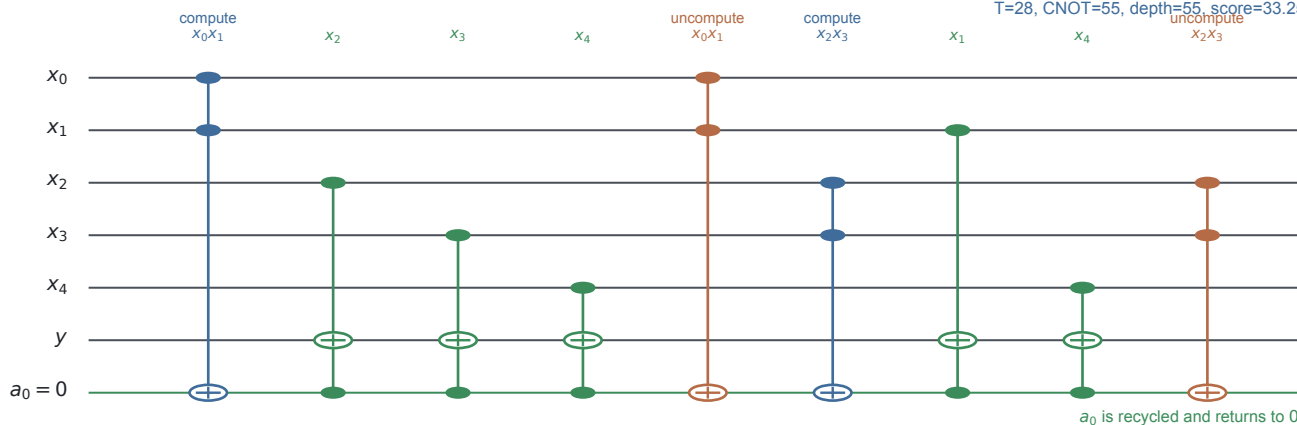
a AND-direct ANF: five independent output actions

T=40, CNOT=65, depth=65, score=45.825



b Resource-NMCTS: two factors share one recycled line

T=28, CNOT=55, depth=55, score=33.255



The selected circuit passes plan expansion, emitted-circuit GF(2) simulation, and all 32 truth-table assignments.